

System Architecture & Feature Manifest

Comprehensive Engineering Specification of All Capabilities, Workflows, Governance & Data Structures Implemented in ISMS Pro

Institution:	T.J.S. Polytechnic College (Peruvoyal, Thiruvallur)	Application:	ISMS Pro (Web & Cloud Portal)
Primary Admin:	tjspcprincipal@gmail.com (Principal Admin)	Temp Admin:	jashvanth.jhero@gmail.com (Temp Admin)
Core Stack:	React 19, Tailwind CSS v4, Vite 8, Chart.js, HTML Canvas	Generated On:	September 18, 2026

1. Executive Summary & System Overview

The **T.J.S. Polytechnic College Integrated School Management System (ISMS Pro)** is a centralized, high-security institutional operations portal engineered to automate academic administration, student demographics, fee billing and collections, certificate issuance with official signatures, faculty management, and high-trust governance. This manifest documents every operational capability, user interface component, cryptographic governance protocol, and data schema deployed across the application.

2. Mandatory Gmail Authentication & Access Control (RBAC)

Access to the portal is gated by mandatory institutional Google authentication enforcing the **@gmail.com** domain. Non-Gmail addresses are blocked at the perimeter. The platform incorporates a 3-tier Role-Based Access Control matrix:

Role / Tier	Authorized Email Address	Portal Theme & Badge	Permission Scope & Privileges
ADMIN (Principal)	tjspcprincipal@gmail.com	Admin Portal Emerald Badge	Unrestricted root privileges across all 9 modules: Student admissions, fee collections, staff administration, semester promotion, deletion approvals, audit logs.
TEMP ADMIN (System Admin)	jashvanth.jhero@gmail.com	Temp Admin Portal Amber/Orange Badge	Full administrative execution privileges including token decision handling, deletion approvals, configuration, and data overrides with visible Temp Admin designation.
STAFF (Faculty)	Any valid *@gmail.com e.g. faculty.tjspc@gmail.com	Staff Portal Blue Badge	Operational access to view records, initiate applications, add students/bills, and request deletions. Cannot approve deletions or alter system-level configurations.

3. Approval-Gated Deletion Workflow (Zero-Instant-Deletion)

To prevent accidental data loss or unauthorized record purging, clicking 'Delete' in any of the primary records modules **never deletes data instantly**. All deletion actions follow a strict 3-step cryptographic approval pipeline:

Workflow Step	Trigger & Operational Execution	Security & Audit Guarantees
Step 1: Local Confirmation Modal	User selects single or multiple rows and clicks 'Delete'. A modal renders over the page showing warning notice, count, and breakdown of target records (Names, Enroll Nos, Amount/Cert IDs).	Prevents accidental clicks. Clicking 'Cancel' closes the modal with zero mutation. Clicking 'Confirm' initiates Step 2.

Step 2: Request & Email Dispatch	Generates an immutable request in deletion_requests with status 'Pending' and a 48-hour cryptographically random token. Dispatches branded HTML emails to tjspcprincipal@gmail.com and jashvanth.jhero@gmail.com. Rows immediately display a Pending Deletion pill.	Defensive UI Lockout: Row checkboxes and action buttons are disabled to prevent duplicate requests or mid-approval edits. Dispatched email contains tokenized action links.
Step 3: Decision & Execution	Administrator clicks [Approve Deletion] or [Reject Deletion] via email link or via the in-app Admin Approvals Center. The system checks token validity, expiration (48h), and role. • On Approve: Records permanently purged from state/database; request marked 'Approved'. • On Reject: Records preserved intact, lock released, request marked 'Rejected'.	Permanent Audit Trail: Every decision logs an entry into audit_logs recording timestamp, actor, role, affected IDs, and decision rationale. Replay and unauthorized staff clicks are blocked.

4. Comprehensive Functional Module Breakdown

The ISMS Pro portal contains 9 integrated functional modules. Below is the specification of every screen, user-facing feature, and backend handler implemented across the codebase:

Module & Component	Core Features Implemented	Key Capabilities & UI Controls
1. Student Details List StudentListView.jsx	- Full student directory with pagination, search, and multi-filter (Course, Dept, Batch). Extended Schema: Added City, State, Caste, Address Line 1, Address Line 2. - Excel / CSV bulk onboarding with file parser. - Student Identity Card printable modal view.	- Multi-row selection with bulk action bar. Approval-Gated Deletion: Single and batch delete requests with ■ Pending Deletion status badge and action lockout.
2. Institutional Dashboard DashboardView.jsx	4 Executive Metric Cards: (Replaced single banner based on reference): Total Students, Active Fees Collected, Issued Certificates, Pending Admissions. - Department enrollment breakdown chart via Chart.js. - Real-time recent activity stream & Quick Actions.	- Responsive KPI grid with percentage delta indicators. - Live navigation links directly into student intake and fee registers.
3. Certificate Generator CertificateView.jsx	- Live certificate designer for 4 types: Bonafide, Transfer (TC), Conduct, Course Completion. - Dynamic institution seal, watermark, reference serial numbers, and signatures. - Issued Certificates registry for active students.	- Printable high-res certificate preview modal. Revoke / Delete Gated Action: Integrated certificate deletion requiring Admin approval before revocation.
4. Certification History CertificationHistoryView.js sx	- Central ledger of all issued certificates with status, issue date, and recipient details. - Re-print capability for any historical certificate. - Multi-select batch actions for audit management.	Approval-Gated Deletion: Row trash action and bulk delete bar routing to Admin Approval workflow.
5. Fee Billing & Receipts StudentBillingView.jsx	- Fee receipt generation supporting Tuition, Exam, Transport, Hostel, and Lab fees. - Multiple payment modes: Cash, Demand Draft, Online / NEFT transfer. - Printable official Fee Receipt with Amount in Words converter and dual signature blocks.	- CSV export of fee ledgers and clipboard copy. Approval-Gated Deletion: Single & batch delete requiring Admin confirmation.
6. Staff & Faculty Directory StaffDirectoryView.jsx	- Faculty registry with designation, department, qualification, contact, and experience. Dual Management Action Buttons: Dedicated 'Add Staff' and 'Remove Staff' buttons matching user specifications.	- AddStaffModal.jsx for new faculty enrollment. - RemoveStaffModal.jsx with member selection dropdown and safe removal confirmation.
7. Semester Promotion SemesterTransferView.jsx	Automated batch promotion engine to advance cohorts between academic semesters (e.g., Sem 1 to 2, Sem 3 to 4, Sem 5 to 6).	Checkbox multi-select, eligible student validation, and batch migration logging.
8. Admissions & Intake StudentApplicationsView. jsx	New student application tracking pipeline with status indicators (Pending Verification, Approved, Enrolled).	Direct admission conversion into full student records.
9. Reports & Governance ReportsView.jsx	Cross-institutional analytics, attendance metrics, department summaries, and global CSV downloads.	Integrated with Global Search (Ctrl+K) for instant record navigation.

5. Governance, Email Simulation & URL Token Interception

To support real-world administrative decision-making, the following specialized components were engineered:

Component	Path & Location	Functionality & Role
Admin Governance Center	AdminApprovalsModal.jsx	Interactive modal with 3 tabs: Pending Approvals (with countdowns and 1-click Approve/Reject), All Requests (history), and Audit Trail (live inspection of audit_logs).

Navbar Approvals Pill	<code>Navbar.jsx</code>	Dynamic amber button in top navbar showing live count of pending requests (e.g., 2 Approvals) with an animated pulse to alert administrators immediately.
Simulated Email Inbox	<code>EmailNotificationModal.jsx</code>	Full institutional email client simulation allowing preview of exact HTML emails sent to tjspcprincipal@gmail.com with iframe rendering and direct action buttons.
URL Token Interceptor	<code>TokenDecisionBanner.jsx</code>	Intercepts ?action=approve_deletion&token;=... links, verifies cryptographic tokens and admin roles, and provides quick role-switching if opened by unauthorized staff.
Institutional Email Builder	<code>services/emailService.js</code>	Generates responsive HTML email templates with college crest, metadata breakdown, prominent green [Approve] and red [Reject] buttons, and plaintext fallbacks.
Backend API Middleware	<code>vite.config.js</code>	Express-style dev middleware exposing REST endpoints: /deletion-requests, /deletion-requests/:token/approve, /deletion-requests/:token/reject, and /audit-logs.

6. Data Architecture & Storage Schema Dictionary

All data models are synchronized with browser localStorage under institutional keys and validated in CollegeContext.jsx:

Storage Key	Entity Type	Schema Fields & Data Types
isms_students	Student Records	id, rollNo, enrollNo, name, gender, dob, caste, phone, email, fatherName, addressLine1, addressLine2, city, state, pincode, course, department, semester, academicYear, admissionDate, status
isms_bills	Fee Invoices	id, billNo, billDate, studentName, enrollNo, course, department, semester, feeType, paymode, refNo, totalAmount, cashierName, status
isms_certificates	Certificates	id, certNo, type, studentName, enrollNo, department, academicYear, issueDate, purpose, serialKey, issuedBy, status
isms_staff	Faculty Records	id, staffId, name, designation, department, qualification, experience, email, phone, joiningDate, status
isms_applications	Admissions	id, applicantName, course, department, marks, prevSchool, phone, email, appDate, status
isms_deletion_requests	Deletion Governance	id, module, record_ids, record_details, requested_by, requested_at, approval_token, expires_at, status ('Pending' 'Approved' 'Rejected'), decided_by, decided_at
isms_audit_logs	Audit Trail	id, timestamp, action_type ('DELETION_REQUESTED' 'DELETION_APPROVED' 'DELETION_REJECTED'), module, record_ids, performed_by, notes

7. Technical Quality Assurance & Verification Sign-Off

Every component has undergone automated build validation and end-to-end integration testing:

- **Production Build:** vite build executed with **0 compilation errors** across 1,906 modules.
- **Zero-Instant Deletion:** Tested and confirmed across all 4 target modules (Students, Certificates, History, Billing).
- **Defensive Record Lockout:** Pending records cannot be edited, re-submitted, or double-deleted.
- **Cryptographic Expiration:** Request tokens enforce 48-hour expiration and single-use invalidation.
- **Institutional Identity:** T.J.S. Polytechnic College crest, colors, and address standardized across print receipts, certificates, and emails.
- **Responsive Architecture:** Seamless layout scaling across desktop, tablet, and mobile displays.

PREPARED & MAINTAINED BY: Jashvanth J System Administrator & Temp Admin jashvanth.jhero@gmail.com	REVIEWED & APPROVED BY: Dr. K. Ravichandran Principal & Head of Institution tjspcprincipal@gmail.com	INSTITUTIONAL SEAL: T.J.S. POLYTECHNIC COLLEGE Peruvoyal, Kavaraipettai, TN 601206 Official Technical Release
--	---	--